

Engineering Tripos Part IIB

Project Planning Form 2026/27

Students should submit this form to Group Administrators by **Friday 12 June 2026**. If plans evolve over the summer, please update and resubmit.

Name of student: Xiancheng Meng

Email: xm272@cam.ac.uk

College: Trinity College

Director of Studies: Per Ola Kristensson
(pok21@cam.ac.uk)

Project is: Type (a)

Name of supervisor: Rich Turner

Short title of project: Adversarial examples in AI Weather Forecasting

Project reference code (e.g. B-ANO22-1 / B-ANO22-type(b)): F-ret26-3

Aims of project:

This project investigates the sensitivity of and how robust AI-based weather forecasting models respond to adversarial perturbations (deliberately chosen changes to the input data that cause the model to make significantly worse or misleading forecasts). More specifically, it will focus on general robustness as well as adversarial vulnerability. We will be likely to start with white-box attacks on Otter Weather, where the attacker has full access to the model and can use gradients or other model information to find worst-case perturbations. It will gradually extend toward transferable and physically sensible attacks on more realistic end-to-end forecasting systems. By comparing adversarial perturbations with random noise, and by studying their effect on extreme-weather forecasts, the project will assess both the robustness and the security risks of modern data-driven weather prediction.

Preparatory work for the long vacation:

1. Reading core papers, learn basic adversarial attack methods: FGSM, PGD, iterative gradient-based attacks.
<https://arxiv.org/abs/2504.20238>
<https://journals.ametsoc.org/view/journals/aies/aop/AIES-D-26-0009.1/AIES-D-26-0009.1.xml>
<https://www.nature.com/articles/s41586-025-08897-0>
<https://www.nature.com/articles/s41586-025-09005-y>
<https://arxiv.org/abs/1810.00069>
<https://arxiv.org/abs/2305.10862>
-
2. Practice implementing simple attacks (potentially using FGSM and PGD) in PyTorch on simple dataset.
3. Revise and learn autoregressive forecasting techniques.
4. Learn common weather forecast variables and metrics: geopotential height, temperature, precipitation.

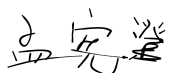
Workshop/laboratory facilities needed:

This project will require access to University HPC (GPU cluster) and CBL high-end desktop and GPU servers.

Any special safety implications:

N/A

Signature of student:



Xiancheng Meng

Signature of supervisor:

